Ordinary Differential Equations

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1 Linear First-Order Equations

A first-order linear differential equation has the form

$$y' + p(t)y = g(t).$$

The goal is to find an integrating factor $\mu(t)$ such that multiplying the entire equation by $\mu(t)$ produces the derivative of a product.

1.1 Integrating Factor

We choose

$$\mu(t) = e^{\int p(t)\,dt}.$$

Multiplying

$$y' + p(t)y = g(t)$$

by $\mu(t)$ gives

$$\mu y' + p(t)\mu y = \mu g.$$

We want the left-hand side to be

$$\frac{d}{dt}(\mu y).$$

By the product rule,

$$\frac{d}{dt}(\mu y) = \mu' y + \mu y'.$$

Therefore, we need

$$\mu' = p(t)\mu.$$

This is satisfied by

$$\mu(t) = e^{\int p(t)\,dt}.$$

Thus,

$$\frac{d}{dt}(\mu y) = \mu g.$$

Integrating both sides gives

$$\mu y = \int \mu(t) g(t)\, dt + C.$$

Therefore,

$$\boxed{y(t) = \frac{1}{\mu(t)} \left(\int \mu(t) g(t)\, dt + C \right).}$$

1.2 Example: Linear Equation

Solve

$$\frac{dy}{dt} + \frac{1}{2} y = t.$$

This is in the form

$$y' + p(t) y = g(t),$$

where

$$p(t) = \frac{1}{2}, \qquad g(t) = t.$$

The integrating factor is

$$\mu(t) = e^{\int \frac{1}{2}\, dt} = e^{t/2}.$$

Multiply the differential equation by $e^{t/2}$:

$$e^{t/2} y' + \frac{1}{2} e^{t/2} y = t e^{t/2}.$$

The left-hand side is the derivative of a product:

$$\frac{d}{dt} \left(e^{t/2} y \right) = t e^{t/2}.$$

Integrate:

$$e^{t/2} y = \int t e^{t/2}\, dt + C.$$

Using integration by parts,

$$\int te^{t/2}\,dt = e^{t/2}(2t-4).$$

Therefore,

$$e^{t/2}y = e^{t/2}(2t-4) + C.$$

Divide by $e^{t/2}$:

$$\boxed{y(t) = 2t - 4 + Ce^{-t/2}}.$$

2 Separable Equations

A first-order differential equation

$$\frac{dy}{dx} = f(x,y)$$

is called **separable** if it can be written in the form

$$M(x)\,dx + N(y)\,dy = 0.$$

Equivalently, we can write

$$N(y)\frac{dy}{dx} = -M(x).$$

The important feature is that the x-dependent terms and y-dependent terms can be separated.

2.1 General Method

Suppose we have

$$M(x)\,dx + N(y)\,dy = 0.$$

Let H_1 and H_2 be antiderivatives satisfying

$$H_1'(x) = M(x)$$

and

$$H_2'(y) = N(y).$$

Then

$$\frac{d}{dx}H_1(x) + \frac{d}{dx}H_2(y) = 0.$$

By the chain rule,

$$\frac{d}{dx}H_2(y) = H_2'(y)\frac{dy}{dx} = N(y)\frac{dy}{dx}.$$

Thus,

$$\frac{d}{dx}\left(H_1(x) + H_2(y)\right) = 0.$$

Therefore,

$$\boxed{H_1(x) + H_2(y) = C}.$$

2.2 Initial Value Problems

Suppose an initial condition

$$y(x_0) = y_0$$

is given.

Since

$$H_1(x) + H_2(y) = C,$$

substituting (x_0, y_0) gives

$$C = H_1(x_0) + H_2(y_0).$$

Therefore, the solution to the IVP can be written as

$$\boxed{H_1(x) + H_2(y) = H_1(x_0) + H_2(y_0)}.$$

This gives the solution implicitly. If possible, we may then solve for y explicitly.

2.3 Example: Separable IVP

Solve the IVP

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}, \qquad y(0) = -1.$$

First separate the variables:

$$2(y-1)\,dy = (3x^2 + 4x + 2)\,dx.$$

Integrate both sides:

$$\int 2(y-1)\,dy = \int (3x^2 + 4x + 2)\,dx.$$

The left-hand side becomes

$$(y-1)^2,$$

while the right-hand side becomes

$$x^3 + 2x^2 + 2x + C.$$

Thus,

$$(y-1)^2 = x^3 + 2x^2 + 2x + C.$$

Use the initial condition

$$y(0) = -1.$$

Then

$$(-1-1)^2 = C,$$

so

$$C = 4.$$

Therefore,

$$(y-1)^2 = x^3 + 2x^2 + 2x + 4.$$

Taking the square root gives

$$y - 1 = \pm\sqrt{x^3 + 2x^2 + 2x + 4}.$$

Since $y(0) = -1$, we must choose the negative branch:

$$\boxed{y(x) = 1 - \sqrt{x^3 + 2x^2 + 2x + 4}}.$$

This is the solution satisfying the initial condition.

3 Homogeneous Equations

A first-order differential equation

$$\frac{dy}{dx} = f(x, y)$$

is called **homogeneous** if the right-hand side can be written as a function of the ratio y/x:

$$\boxed{\frac{dy}{dx} = g\left(\frac{y}{x}\right)}.$$

The important observation is that a homogeneous equation can be converted into a separable equation using the substitution

$$\boxed{v = \frac{y}{x}}.$$

Equivalently,

$$y = vx.$$

3.1 The Homogeneous Substitution

Starting with

$$y = vx,$$

differentiate both sides with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}.$$

Therefore, whenever

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right),$$

we substitute

$$\frac{y}{x} = v$$

and

$$\frac{dy}{dx} = v + x\frac{dv}{dx}.$$

This gives

$$v + x\frac{dv}{dx} = g(v).$$

Rearranging,

$$x\frac{dv}{dx} = g(v) - v.$$

Hence,

$$\frac{dv}{g(v) - v} = \frac{dx}{x}.$$

The equation is now separable.

3.2 Example: Homogeneous Equation

Solve

$$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2}.$$

First rewrite the right-hand side:

$$\frac{x^2 + xy + y^2}{x^2} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2.$$

Therefore,

$$\frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2.$$

This is homogeneous because the right-hand side depends only on y/x.

Use the substitution

$$v = \frac{y}{x},$$

so that

$$y = vx.$$

Differentiate:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}.$$

Substitute into the differential equation:

$$v + x\frac{dv}{dx} = 1 + v + v^2.$$

Cancel v from both sides:

$$x\frac{dv}{dx} = 1 + v^2.$$

Separate the variables:

$$\frac{dv}{1+v^2} = \frac{dx}{x}.$$

Integrate:

$$\int \frac{dv}{1+v^2} = \int \frac{dx}{x}.$$

Therefore,

$$\arctan(v) = \ln|x| + C.$$

Substitute back

$$v = \frac{y}{x}:$$

$$\arctan\left(\frac{y}{x}\right) = \ln|x| + C.$$

This is an implicit form of the solution.

Solving for y/x,

$$\frac{y}{x} = \tan(\ln|x| + C).$$

Thus,

$$\boxed{y = x\tan(\ln|x| + C)}.$$

4 Summary of Methods

4.1 Linear First-Order Equations

For

$$y' + p(t)y = g(t),$$

use the integrating factor

$$\boxed{\mu(t) = e^{\int p(t)\,dt}}.$$

Then

$$\boxed{\frac{d}{dt}(\mu y) = \mu g}$$

and therefore

$$\boxed{y = \frac{1}{\mu}\left(\int \mu g\,dt + C\right)}.$$

4.2 Separable Equations

If an equation can be written as

$$M(x)\,dx + N(y)\,dy = 0,$$

then integrate:

$$\boxed{\int M(x)\,dx + \int N(y)\,dy = C}.$$

For an IVP

$$y(x_0) = y_0,$$

use the initial condition to determine C.

4.3 Homogeneous Equations

If

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right),$$

use

$$\boxed{v = \frac{y}{x}}$$

so that

$$\boxed{y = vx}$$

and

$$\boxed{\frac{dy}{dx} = v + x\frac{dv}{dx}}.$$

The resulting equation should be separable.